Relative Motion Problem

(Rain and Man)

1. A man while walking on a horizontal road with a velocity of 10 km/h observes the rains to be falling vertically downward. If he increases his speed to 20 km/h, then the rain drops appear to strike him at an angle of $45°$ from vertical. What is then actual velocity of.the rain drops ?

   Ans. $10\sqrt{2}$ km/h at angle of $45°$ from vertical.

2. Rain appears to fall vertically to a man when he is stationary but when he starts moving at speed of 10 km/h; the rain drops appears to strike him at an angle of $45°$ from the vertical, then determine the actual velocity of the rain drops.

   Ans. 10 km/h.

3. Rain appears to fall vertically to a man when he is stationary but when he starts moving at speed of 10 km/h; the rain drops appears to strike him at an angle of $45°$ from the vertical, then determine the actual velocity of the rain drops.

   Ans. 10 km/h.

4. Rain drops appears to a man falling vertically. But when he starts moving with a velocity of 10 m/sec horizontally, then the rain drops appears to hit him at an angle of $60°$ from horizontal. Then what is the actual velocity of the rain drops ?

   Ans. $10\sqrt{3}$ m/sec.

5. A school going girl is running on a horizontal road with a velocity of 4 m/sec. At what angle she should hold her umbrella in order to protect herself from the rains. If the rain is falling vertically with a velocity 8 m/sec.

   Ans. $\theta = \tan^{-1}(2)$ from the horizontal

6. Rain appears to fall vertically downward to a man moving with a velocity of 4 m/sec. when he doubles his speed then the rain appears to strike him at an angle of $45°$ from horizontal. Then what is the actual velocity of rain drops ?

   Ans. $4\sqrt{2}$ m/sec, at angle of $45°$ from vertical.

7. Rain appears to fall vertically downward to a man moving with a velocity of 10 m/sec, when he doubles his speed then the rain appears to strike him at an angle of $60°$ from horizontal. then what is actual velocity of the rain drops ?

   Ans. 20 m/sec at angle of $30°$ from vertical.

8. A stationary man observes that the rain strikes him at an angle of $60°$ to the horizontal. When he begins to move with a velocity of 25 m/sec, then the drops appear to strike him at an angle $30°$ from horizontal. Calculate the velocity of the rain drops.

   Ans. 25 m/sec.

9. A boy is running on the plane road with velocity (v) with a long hollow tube in this hand. The water is falling vertically downwards with velocity (u). At what angle to the vertical, he must incline the tube so that the water drops enters in it without touching its side –

   (A) $\tan^{-1}\left(\frac{v}{u}\right)$ (B) $\sin^{-1}\left(\frac{v}{u}\right)$

   (C) $\tan^{-1}\left(\frac{u}{v}\right)$ (D) $\cos^{-1}\left(\frac{v}{u}\right)$